

### EXAMPLE 3.4 Magical Gaussian Quadrature

The energy density of blackbody radiation at temperature  $T$  is given by

$$u(T) = \frac{8\pi(kT)^4}{(hc)^3} J \quad (3.24)$$

where  $h$ ,  $c$ , and  $k$  are Plank constant[1], speed of light, and Boltzmann constant[2]. The factor  $J$  is a dimensionless constant determined by integral:

$$J = \int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15} \quad (3.25)$$

Let us try to integrate it numerically and compare the result with the exact value. Since the integral bounds are 0 and  $\infty$ , we use the Gaussian-Laguerre quadrature. To use the Gaussian-Laguerre quadrature, the integrand must have  $e^{-x}$ . We can always create  $e^{-x}$  by multiplying  $e^x e^{-x}$  to the integrand. Now we have the desired exponential function  $e^{-x}$ , but as a penalty another exponential function  $e^x$  is introduced in the integrand which is now  $f(x) = \frac{x^3 e^x}{e^x - 1}$ . We need to make it sure that the extra  $e^x$  does not cause a problem. Since  $f(x) \rightarrow x^3$  as  $x \rightarrow \infty$ ,  $f(x)$  behaves like a polynomial and thus the Gaussian quadrature is expected to give a good result. Program 3.3 calculate  $J$  using the Gauss-Laguerre quadrature. The output is

```
8 points Gaussian Laguerre Quadrature
Exact= 6.493939402267e+00
Gauss= 6.493935665353e+00
Error= 3.736914144348e-06
```

Despite that we evaluated the integrand only at 8 points, the agreement with the exact value is remarkable.

## 3.6 Applications in Physics

### 3.6.1 Period of Classical Oscillation I.

A classical particle with energy  $E$  is confined in a potential  $U(x)$ . (See Fig 3.4.) The particle oscillates between turning points  $x_1$  and  $x_2$ . The period of oscillation[3] is given by

$$T = 2 \int_{x_1}^{x_2} \frac{1}{v(x)} dx \quad (3.26)$$

where the speed of the particle at  $x$  is given by

$$v(x) = \sqrt{\frac{2(E - U(x))}{m}} \quad (3.27)$$

The integral bounds are determined by solving  $v(x) = 0$ , which requires numerical root finding discussed in next Chapter.

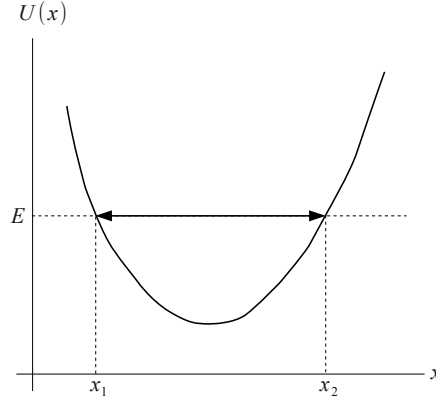


Figure 3.4: Classical Oscillation

The integral in Eq. (3.26) is improper since  $v(x_1) = v(x_2) = 0$ . We need to remove the integral singularities. First we expand the potential around the turning points:

$$U(x) = U(x_i) + U'(x_i)(x - x_i) + o(x^2) = E + U'(x_i)(x - x_i) + o(x^2), \quad i = 1, 2 \quad (3.28)$$

If the potential is approximated by  $U_i(x) = E + U'(x_i)(x - x_i)$ , the speed becomes

$$v_i(x) = \sqrt{\frac{-2U'(x_i)(x - x_i)}{m}} \quad (3.29)$$

Note that this approximated speed approaches to the correct speed as  $x \rightarrow x_i$ . Utilizing it, we remove the singularities as follows.

$$\int_{x_1}^{x_2} \frac{1}{v(x)} dx = \int_{x_1}^{x_0} \left[ \frac{1}{v(x)} - \frac{1}{v_1(x)} \right] dx + \int_{x_1}^{x_0} \frac{1}{v_1(x)} dx \quad (3.30)$$

$$+ \int_{x_0}^{x_2} \left[ \frac{1}{v(x)} - \frac{1}{v_2(x)} \right] dx + \int_{x_0}^{x_2} \frac{1}{v_2(x)} dx \quad (3.31)$$

$$(3.32)$$

where  $x_0$  is any point between  $x_1$  and  $x_2$ . A good choice would be a point where the potential is minimum. The integrands inside the square brackets have no singularity and thus can be integrated using a common numerical method. The remaining integral can be easily evaluated analytically:

$$\int_{x_1}^{x_0} \frac{1}{v_1(x)} dx = \sqrt{\frac{m}{-2U'(x_1)}} \int_{x_1}^{x_0} \frac{1}{\sqrt{x - x_1}} dx = \sqrt{\frac{2mx_0}{|U'(x_1)|}} \quad (3.33)$$

$$\int_{x_0}^{x_2} \frac{1}{v_2(x)} dx = \sqrt{\frac{m}{2U'(x_2)}} \int_{x_0}^{x_2} \frac{1}{\sqrt{x_2 - x}} dx = \sqrt{\frac{2mx_0}{|U'(x_2)|}} \quad (3.34)$$

### 3.6.2 Scattering by Yukawa Potential: Part 1

A particle of mass  $m_1$  elastically collides with another particle of mass  $m_2$  through a spherical potential  $U(r)$  where  $r$  is the distance between the particles. The scattering angle  $\theta$  defined in Fig. 3.5 depends on

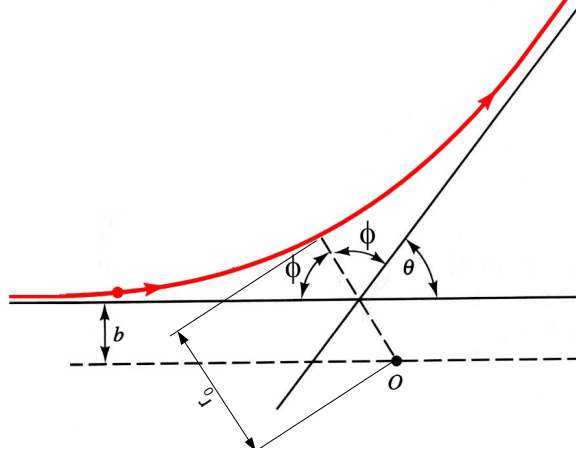


Figure 3.5: Geometry of scattering in relative coordinate.

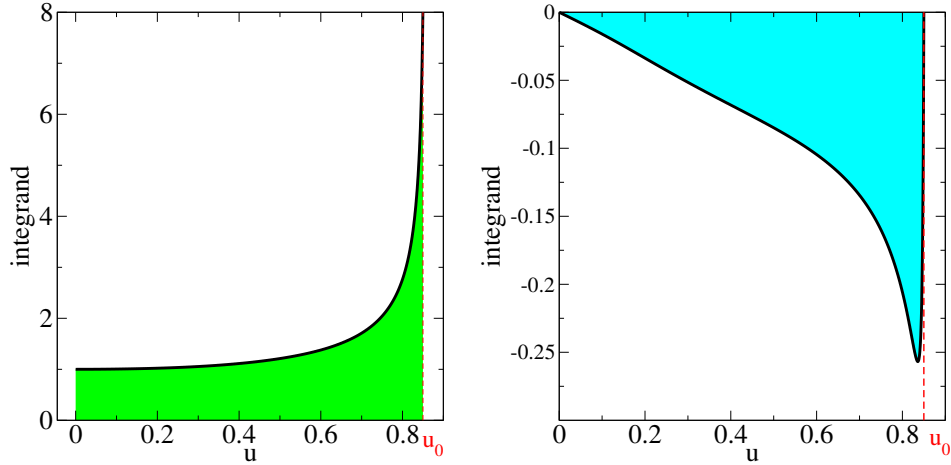


Figure 3.6: The left panel shows the original integrand. The green area need to be numerically integrated. The right panel shows the integrand after the singularity is removed. The blue area need to be integrated. Note the difference in scale between two plots. The blue area is much smaller than the green area. Parameter values  $k = a = E = 1$  are used.

the impact parameter  $b$  and the energy of the system  $E$ . For mathematical convenience, we introduce a new angle  $\phi$  as shown in Fig. 3.5. Note that  $\theta = \pi - 2\phi$ . This scattering problem can be analytically solved up to the following integral[4]:

$$\phi = \int_{r_0}^{\infty} \frac{b}{r^2} \frac{1}{\sqrt{1 - \frac{b^2}{r^2} - \frac{U(r)}{E}}} dr . \quad (3.35)$$

The lower integral limit  $r_0$  is the closest distance between two particles determined by the equation

$$1 - \frac{b^2}{r_0^2} - \frac{U(r_0)}{E} = 0 . \quad (3.36)$$

For the Coulomb potential this integral can be analytically carried out. (Rutherford scattering). [4] We want to find the scattering angle for Yukawa potential (screened Coulomb potential):

$$U(r) = \frac{k}{r} e^{-r/a}$$

where  $k$  and  $a > 0$  are constant.

This integral is improper in two reasons. One is that the upper integral limit is infinity. The other is that the integrand diverges at  $r = r_0$ . See the singularity in Fig. 3.6. The first difficulty can be easily resolved. Introducing a variable  $u = \frac{1}{r}$ . Eqs. (3.35) and (3.36) are respectively transformed to

$$\phi = \int_0^{u_0} \frac{b}{\sqrt{1 - b^2 u^2 - \frac{U(1/u)}{E}}} du \quad (3.37)$$

and

$$1 - b^2 u_0^2 - \frac{U(1/u_0)}{E} = 0. \quad (3.38)$$

Removing the singularity is a bit more difficult. The method used in Example 3.3 is helpful. Noting that the Rutherford scattering can be solved analytically, we consider scattering by Coulomb potential  $U_C(r) = \frac{ke^{-r_0/a}}{r}$ . Expressing in variable  $u$ , the scattering angle by  $U_C$  is analytically obtained:

$$\phi_C = \int_0^{u_0} du \frac{b}{\sqrt{1 - b^2 u^2 - c u}} \quad (3.39)$$

$$= \sin^{-1} \left( \frac{2b^2 u_0 + c}{\sqrt{c^2 + 4b^2}} \right) - \sin^{-1} \left( \frac{c}{\sqrt{c^2 + 4b^2}} \right). \quad (3.40)$$

where  $c = \frac{ke^{-r_0/a}}{E}$ . Now the scattering angle by the Yukawa potential is given by  $\phi = \phi_C + \Delta\phi$  where

$$\Delta\phi = \phi - \phi_C \quad (3.41)$$

$$= \int_0^{u_0} du \left[ \frac{b}{\sqrt{1 - b^2 u^2 - c u e^{-(1/u - 1/u_0)/a}}} - \frac{b}{\sqrt{1 - b^2 u^2 - c u}} \right] \quad (3.42)$$

The integrand in the square bracket is no longer singular at  $u = u_0$  because the singularity in the two terms is exactly canceled. Hence,  $\Delta\phi$  can be evaluated by a standard numerical integral. Figure 3.6 shows that the integrand of Eq. (3.42) does not have the divergence any more. We can use simple numerical integration algorithms such as the trapezoidal or the Simpson rule to integrate Eq. (3.42). One issue is that the integrand has sharp change near  $u_0$ . Try  $N=100, 500, 1000$  in the Simpson rules. If the results do not change significantly, you have sufficiently accurate results.\*

We are still not ready to write a program yet. We must find  $r_0$  by solving Eq. (3.36) or (3.38). Unfortunately, there is no analytical solution for the Yukawa potential. We need a numerical root finding method, which we will discuss in next chapter.

### 3.6.3 Debye Model of Heat Capacity

Based on Debye theory, the heat capacity of a solid at temperature  $T$  is given by

$$C_V = 9k_B N \left( \frac{T}{\theta_D} \right)^3 \int_0^{\theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx \quad (3.43)$$

\*We investigate the same problem with a different method in Chapter 4

where  $\theta_D$  is the Debye temperature,  $N$  is the number of atoms, and  $k_B$  is Boltzmann's constant.[5] Since the upper limit of the integral depends on temperature, we may need to use different numerical methods for different temperature.

(1) Before going to calculation, first we normalize quantities. The heat capacity of ideal gas consisting of  $N$  particles is  $\frac{3}{2}k_B N$ . Using this as a unit of heat capacity, the heat capacity of the material is  $\tilde{C}_V = 2C_v/3k_B N$ . We measure temperature using the Debye temperature as unit. The normalized temperature is  $\tilde{T} = T/\theta_D$ . Now, the original expression is written as

$$\tilde{C}_V = 6\tilde{T}^3 \int_0^{1/\tilde{T}} \frac{x^4 e^x}{(e^x - 1)^2} dx \quad (3.44)$$

which does not include very large number. More importantly, this expression does not depend on the actual material (neither  $N$  nor  $\theta_D$ ). Therefore, the result is universal.

(2) When the temperature is similar or larger than the Debye temperature ( $\tilde{T} \gtrsim 1$ ), the upper limit of the integral is about 1 or smaller. There is no numerical difficulty to integrate it numerically. The Simpson or trapezoidal rule is sufficient.

(3) When the temperature is exactly zero, the upper limit of the integral is infinity. As  $x \rightarrow \infty$ , the integrand behaves as  $x^4 e^{-x}$ . Hence, the integral is finite and the Gaussian-Laguerre quadrature should work well. However, it is not necessary to calculate the integral since the factor  $\tilde{T}^3$  in front of the integral vanishes. Thus,  $\tilde{C}_V = 0$ .

(4) When the temperature is much lower than Debye temperature but still above zero, the upper limit of the integrals can very large. The direct integration using the Simpson rule may fail. As we learned in part (3), the Gaussian quadrature works well if the upper limit is infinity. Utilizing it we split the integral in two parts.

$$\int_0^{1/\tilde{T}} \frac{x^4 e^x}{(e^x - 1)^2} dx = \int_0^\infty \frac{x^4 e^x}{(e^x - 1)^2} dx - \int_{1/\tilde{T}}^\infty \frac{x^4 e^x}{(e^x - 1)^2} dx \quad (3.45)$$

$$= \frac{4\pi^4}{15} - \int_0^{\tilde{T}} \frac{e^{1/s}}{s^6 (e^{1/s} - 1)^2} ds \quad (3.46)$$

where  $s = 1/x$  is used in the second line. The first integral in the lhs is analytically calculated. In the second integral, the upper limit is  $\tilde{T} \ll 1$  and the Simpson rules is sufficient. Note that when  $s \rightarrow 0$ , the integrand vanishes.

### 3.6.4 Heat Capacity of Free Electron Gas

The heat capacity of free electron gas is given by

$$C_e = \frac{3}{2}k_B N \frac{k_B T}{\varepsilon_F} \int_{-\varepsilon_F/k_B T}^\infty \frac{x^2 e^x}{(e^x + 1)^2} dx \quad (3.47)$$

where  $\varepsilon_F$  is Fermi energy.[6] See Section 3.6.3 for the meaning of other symbols. Expressing the above equation using the normalized heat capacity  $\tilde{C} = 2C/3k_B N$  and  $\tilde{T} = k_B T/\varepsilon_F$ . (see Section 3.6.3), Equation (3.47) is simplified to

$$\frac{\tilde{C}}{\tilde{T}} = \int_{-1/\tilde{T}}^\infty \frac{x^2 e^x}{(e^x + 1)^2} dx \quad (3.48)$$

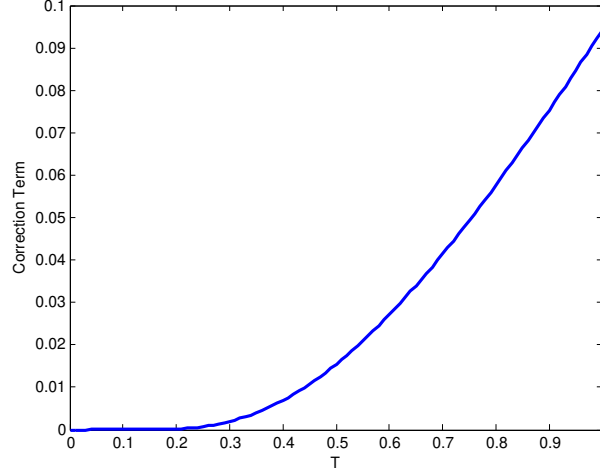


Figure 3.7: Correction term, the integral in Eq. (3.50).

This integral is simpler than Eq. (3.44) since there is no singularity. On the other hand, the upper bound is infinity. For a typical metal at a room temperature or lower,  $\tilde{T} \ll 1$  and thus the lower limit is often replaced with  $-\infty$ . Then, we find an analytical solution

$$\frac{\tilde{C}}{\tilde{T}} \approx \int_{-\infty}^{\infty} \frac{x^2 e^x}{(e^x + 1)^2} dx = \frac{\pi^2}{3} \quad (3.49)$$

If we want to know the heat capacity at a higher temperature we need to integrate the original equation numerically. However, we don't have to evaluate the whole integral. We just need to find the deviation from the low temperature limit (3.49).

$$\begin{aligned} \frac{\tilde{C}}{\tilde{T}} &= \frac{\pi^2}{3} - \int_{-\infty}^{-1/\tilde{T}} \frac{x^2 e^x}{(e^x + 1)^2} dx \\ &= \frac{\pi^2}{3} - \int_{-\infty}^{-1/\tilde{T}} \frac{x^2 e^x}{(e^x + 1)^2} dx \\ &= \frac{\pi^2}{3} + \int_0^{\tilde{T}} \frac{e^{1/s}}{(e^{1/s} + 1)^2} ds \\ &= \frac{\pi^2}{3} + \int_0^{\tilde{T}} \frac{e^{-1/s}}{(e^{-1/s} + 1)^2} ds \end{aligned} \quad (3.50)$$

In the last line,  $\exp(-2/s)$  is multiplied to both the numerator and denominator so that no large number appears. The integral in the last line can be estimated by a standard numerical method.

Figure 3.47 shows that the correction becomes significant only for  $\tilde{T} > 0.2$ . The Fermi energy of a typical material is about  $1\text{ eV}$  to  $10\text{ eV}$ , which corresponds to temperature  $10^4\text{ K}$  to  $10^5\text{ K}$ . Hence, the correction term is not necessary until  $T = 2000\text{ K}$  to  $20000\text{ K}$ . At this temperature, the metal is melt. Thus, the correction term may be safely ignored.

**Problems**

- 3.1** Integrate  $x \cos x$  from  $x = 0$  to  $x = \pi$  using Trapezoidal and Simpson rules. Compare the results with the exact solution  $\int x \cos x \, dx = \cos x + x \sin x$ .
- 3.2** Plot the molar heat capacity of copper ( $\theta_D = 309 \, K$ ) from  $T = 0 \, K$  to  $T = 1083 \, K$  using the Debye theory shown in Section 3.6.3.
- 3.3** Write a code to produce Fig. 3.7.